



Your bridge to every conversation

A real-time translation app for chat, voice, and video — each person speaks or types in their own language, and hears the other in theirs.

Built with AssemblyAI · AssemblyAI Hackathon 2026

Every word in OHUN passes through AssemblyAI first.

Not a bolt-on feature — AssemblyAI is the first stop for everything said in the app, on a call or in a voice note. Three of its capabilities specifically, used deliberately rather than defaulted into:

1

Universal-Streaming

Realtime, turn-by-turn transcription on every call — low enough latency that a translated conversation still feels like talking, not waiting.

2

Batch transcription

Voice notes use AssemblyAI's batch API instead — broader language coverage than realtime streaming, which is how Yorùbá works in chat today.

3

Confidence, tuned

`language_confidence_threshold` set explicitly — a low-confidence guess fails loudly instead of silently mistranscribing.

THE PROBLEM

Language still gets in the way of a real conversation.

Translation apps break the flow

Pass a phone back and forth, wait for typing, read instead of talk. It's a workaround, not a conversation.

Human interpreters don't scale

Expensive, scheduled in advance, and simply not there for a casual call or a quick message.

Group settings are worse

Three or more languages in one room and every existing option collapses immediately.

THE SOLUTION

OHUN turns a translated conversation into an ordinary one.

Speak or type in your language. The other person reads or hears it, live, in theirs. No app switching, no waiting on a human, no one carrying the language burden alone — AssemblyAI hears it, Groq carries the meaning across.

Live translation

Voice & video calls

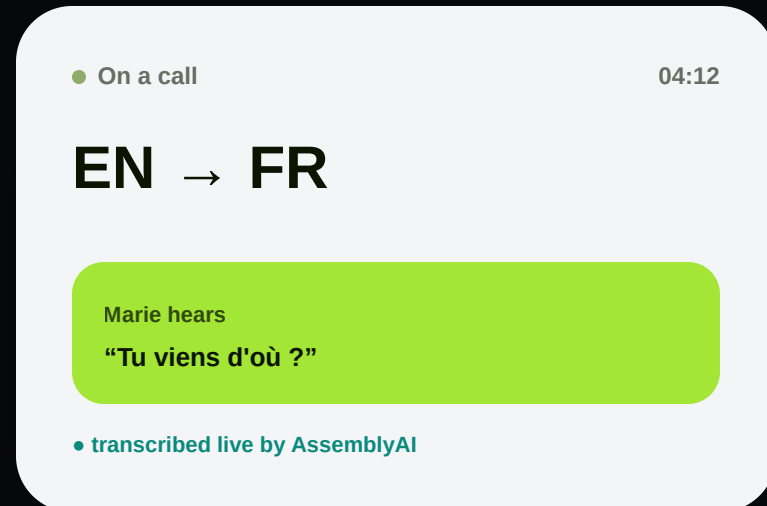
Group calls

Chat & voice notes

Room links

HOW IT WORKS

Three steps, every time — chat, calls, or a group room.



1

You speak AssemblyAI

Universal-Streaming transcribes your side turn by turn, in your own language, as you talk — the foundation everything downstream depends on.

2

Groq translates

Each finished sentence is translated for meaning, not word for word, into their language — gpt-oss-120b on Groq's low-latency inference.

3

They hear it

Spoken aloud on their device with the Web Speech API, and captioned for both sides.

WHAT IT DOES

One account. Every way people actually talk.

Translated chat & voice notes

Message anyone with an OHUN account — every text and voice note arrives already translated, both ways.

Voice & video calls, 1:1

Live speech translation over a real call, captioned and spoken aloud, with camera or screen share added mid-call.

Group calls, up to 7

Any mix of supported languages in one room — everyone hears everyone else translated into their own.

Room links

A personal, shareable call link. Anyone can reach you and start a translated call, no account required.

Scheduling

Book a call for later; the other person gets an email invite with the details.

Call summaries

A short recap after each call, written by the model on Groq — one version per language present.

UNDER THE HOOD

AssemblyAI first, then everything else.

1

Short-lived token, minted server-side

The browser never sees the AssemblyAI API key — it gets a scoped, short-lived streaming token per call and streams audio to AssemblyAI directly.

2

Peer-to-peer media

Supabase Realtime carries only SDP/ICE signalling. STUN plus a Metered TURN relay when a direct connection can't form.

3

Translation stays server-mediated

Transcript text (never audio) routes browser → server → Groq → browser for translation.

Translation loop

mic / typed message

→ AssemblyAI Universal-Streaming
(realtime transcription, turn by turn)

→ Groq (gpt-oss-120b)
(meaning-preserving translation)

→ captioned + spoken aloud

TECH STACK

Built on infrastructure that scales with it.

AI

AssemblyAI

Universal-Streaming for realtime, turn-by-turn call transcription; batch transcription for voice notes, with a broader language footprint than streaming alone — including Yorùbá. Every spoken word in the product starts here.

Next.js 16 + React 19

App Router, Server Actions, Turbopack.

Supabase

Postgres with Row Level Security, Auth, Realtime, Storage.

Groq

gpt-oss-120b for meaning-preserving translation and post-call summaries, fast enough for a live call.

WebRTC

RTCPeerConnection directly — no external calling SDK — with STUN + Metered TURN.

Web Push + Tailwind CSS 4

Installable PWA with push notifications, including on iOS Safari.

LANGUAGES

Starting small. Built to grow — with AssemblyAI's coverage.

English, French, Spanish, German, Portuguese, Italian, and Yorùbá — every one of them works for chat and voice notes, transcribed by AssemblyAI's batch API. Calls run on the six AssemblyAI's Universal-Streaming model handles in realtime today; Yorùbá joins calls the moment a realtime model covers it.

 English

 Français

 Español

 Deutsch

 Português

 Italiano

 Yorùbá · batch API

ROADMAP

Where it goes from here.

1

Yorùbá on calls

Waiting on — or partnering toward — a realtime AssemblyAI model for Yorùbá, closing the one gap between chat and calls.

2

More languages

Grow the supported list past seven as AssemblyAI's coverage and translation quality are verified for each.

3

Larger group calls

Move past the mesh topology to a selective forwarding unit to scale beyond 7 participants.

THANK YOU

Free while we build.

Built with AssemblyAI.

OHUN is in preview and calls are unlimited. Paid plans arrive once the core experience is finished.

github.com/TeevincsCrypt/ohun